311-213 Respiratory system

ข้อสอบ MCQ

1.ข้อสอบมีทั้งหมด 65 ข้อ (ข้อสอบจริงมีจำนวน 100 ข้อ)

2.นักศึกษามีเวลาทำข้อสอบ 150 นาที

3.ข้อสอบจริงเป็นภาษาอังกฤษทั้งหมด

หมายเหตุ : ข้อสอบฉบับนี้เกิดจากความร่วมมือของคนในรุ่นและฝ่ายวิชาการ MED รุ่น 49

1. A doctor intends to ligate an artery, but he accidentally ligates phrenic nerve instead. What does he intend to ligate actually?

a. Musculophrenic artery
b. Internal thoracic artery
c. Intercostal artery
d. Superior epigastric artery
e. Pericardiacophrenic artery

2. A 45-year-old woman stridor found vocal cord adduction อาการดังกล่าวเกี่ยวกับส่วนใด

a. Cricothyroid
b. Cricoarytenoid
c. Epiarytenoid
d. Oblique arytenoid
e. Posterior cricoarytenoid

3. 10-year-old boy were stabbed manubriosternal. Which organ will be affected?

a. Thyroid gland
b. Brachiocephalic trunk
c. Azygos vein

d. tracheal bifurcutated (Med44, Med47)

4. A 3-year-old man suspected a small metal bottom while sitting upright position and play with his toys. Which of the following is most likely site of this falling bottom?

a. Lingula
b. Right upper lobe
c. Left upper lobe
d. Right lower lobe
e. Left lower lobe

ด้านขวา ↓ เป็นแนวตั้ง → มีของลงไปติดง่าย

5. บริเวณไหน lack of smooth muscle

a. Trachea → มี trachealis muscle
b. Bronchus
c. Bronchioles
} smooth muscle มากแถวๆ cartilage
d. Alveolar duct → มี Alveolar knob (มี smooth muscle)
e. Alveolus

6. Which of the following non-ciliated cuboid epithelium associated in secreting product?

a. Clara cell — columnar epi → dome shape → หลั่ง glycoprotein
b. Cell type1 ✗
c. Goblet cell — Columnar
d. Basal cell ✗

Cell type 2 ↳ cuboidal cell → หลั่ง surfactant

7. ข้อใดสามารถเพิ่ม lung compliance ได้ → ความสามารถในการยืดขยาย ↑

a. increase alveola tension
b. increase lung elastic
c. decrease alveoli surface → surface tension ↓
d. decrease surfactant
e. decrease chest wall elastic

surface tension
elastin & collagen
แรงต้านการขยาย

C_L ↑ elastic force ↓

$C_L = \frac{\Delta V_L}{\Delta P_{TP}}$

C_L ↓ elastic ↑

8. Which of the following conditions reduce of lung perfusion zone 1?

a. supine position

↳ active expiration

9. Which following of the volume gas in the lung after abdominal muscle contract

a. RV
b. TLC
c. IRV
d. ERV
e. FRC

10. Normal subject has respiratory rate 10 times/min, total ventilation 8 L/min and anatomical dead space 200 ml. What is his alveolar ventilation?

a. 7.2
b. 7.8
c. 6.0
d. 8.2
e. 0.72

$V_A = (V_T - V_D) \times f$

$= V_T f - V_D f$

$= 8 - (0.2)(10)$

$= 6$

11. Which of the following is higher in emphysema patient compared to normal healthy person?

↳ elasticity ↓

RV ↑ , FRC ↑

↳ อากาศเหลือค้าง ↑

a. FEV1
b. FVC
c. Airway diameter
d. Vital capacity
e. Residual volume

13. What mechanism of pulmonary edema of pneumonia?

a. Increase oncotic plasma pressure
b. Increase vascular permeability
c. Increase hydrostatic plasma pressure

d. Decrease plasma oncotic

e. Decrease lymphatic drainage of visceral pleura

14. When the V/Q ratio of the lung decreased, what alveolar gas tension would expect?

a. Decreased PN2

b. Increased PCO2

c. Increased both PN2 and PCO2

d. Increased both PO2 and PCO2

e. Increased PO2

15. Barometer 525 mmHg ถาม subject's PaO2

ตอบ $PAO_2 = F_iO_2 (P_{atm} - P_{H_2O})$

$= 0.21 (525 - 47 \text{ mmHg}) = 100 \text{ mmHg}$

16. ทำไม PO2 น้อยกว่า PO2 atmospheric dead air

ตอบ → alveolar

O_2 rapidly diffuses into the pulmonary capillary

17. ปัจจัยที่ทำให้ Increase PCO2 ในคนไข้ asthma

a. respiratory membrane thickness

b. surface area ลด

c. ผลต่างความดันระหว่างถุงลมและหลอดเลือดน้อยลง

18. Increase arterial PCO2 ในคนสุขภาพดี

a. moderate exercise

b. mountain climbing ออกกำลังหนักๆ → ↓ P_{O_2}

c. scuba diving

19. What factor CO diffusion limit?

a. Blood pressure

surface area

tissue thickness

20. Deep and slow breathing pattern would be occurred in person who has severe injury brainstem in which brain area?

a. Medulla oblongata
b. Upper pons
c. Thalamus
d. Midbrain
e. Cerebellum

21. Which of the following factor stimulates chemoreceptor resulted in hyperventilation in diabetes mellitus?

a. decreased PO2 in arterial blood
b. increased [H+] in arterial blood
c. increased PO2 in venous blood
d. decreased PCO2 in venous blood
e. increased [H+] in cerebrospinal fluid

22. Which of the following mechanisms is responsible of an increase ventilation during first minute of an aerobic exercise?

a. activation proprioceptor
b. activation peripheral receptor

23. เด็ก 2 ปี มีไข้ ไอ running nose, wheezing โจทย์ถามว่ายาอะไรที่รักษาได้เร็ว Onset เร็ว

a. Inhaled albuterol → 5 min
b. Inhaled cromolyn
c. Oral montelukast
d. Intravenous theophylline
e. Inhaled beclomethasone

2-4 week
1-3 h.
30-60 min
> 1 h.

24. Which of the following medication is inhibit effect block CysLTs by competitive inhibitors to their receptor?

ตอบ Montelukast

25. ถาม MOA ของยา chlorpheniramine

a. Alpha-1 adrenergic agonist
b. H1 receptor
c. H2 receptor

26. 33-weeks-neonatal periodic of breathing association with bradycardia, dx. apnea of prematurity ถามยาที่ใช้ในข้อนี้

a. Zileuton - ASTHMA
b. Albuterol - BRONCHO SPASM, COPD
c. Theophylline - BRONCHODIALATOR
d. Omalizumab - ⊖ IgE

27. A 4-year-old boy is diagnosed acute bronchitis with sticky sputum. Which of the following mucolytic drug is the most appropriate in this patient?

a. Bromhexine - ↓ VISCOSITY OF SPUTUM → ↑ MUCUS CLEARANCE
b. Benzonatate - ⊖ COUGH REFLEX
c. Dextromethorphan ⊖ COUGH REFLEX
d. Loratadine ⊖ H_1 receptor
e. Codeine ↑ SIDE EFFECTS

28. ให้อาการของ TB มา พบ granuloma ถามว่าจะพบอะไรได้อีก

a. พบ Langhans giant cell
b. พบ keratin pearl

29. A 25-year-old man develops fever, cough, dyspnea for a week. Chest x-ray patchy infiltration of right lung. Histopathology of lung tissue demonstrates granulomatous inflammation and mucicarmine stain reveal yeast-like organisms with budding and pink color mucin staining around the organisms. What is diagnosis of this lesion?

a. penicilliosis
b. tuberculosis
c. candidiasis
d. histoplasmosis
e. cryptococcosis

30. Which of the following is the common abnormality associated with diaphragmatic hernia?

a. Lung hypoplasia
b. Pulmonary agenesis
c. Intrauterine infection
d. Congenital heart disease
e. Hyaline membrane disease

Hernia -> ⬆️ compression in lungs -> lung hypoplasia

31. 50-year-old kidney transplantation 3 สัปดาห์ต่อมา มีไข้ ไอ dyspnea bronchoscopy เจอ bronchial epithelial cell enlargement with intracytoplasmic inclusion ข้อไหนคือสาเหตุของอาการดังกล่าว

a. Adenocarcinoma
b. CMV
c. HSV
d. RSV

32. หญิง 25 ปีเป็น kindergarten teacher ไปโรงพยาบาลด้วย 2 days history of fever, sore throat, sore clearing, postnasal drip, rhinorrhea, and nasal stuffiness. What is the most common of this URI.

a. Influenza virus
b. Rhinovirus

c. human metapneumovirus

d. RSV

e. Streptococcus pneumonia

33. 50-year-old female presented with cough, and right-sided chest pain for 15 days. Gram sputum revealed numerous neutrophils and a rare amount of Gram-positive thin branching beaded filamentous bacteria; however, the result of Kinyoun staining were negative.

What is the most likely negative organism?

a. Mycobacterium tuberculosis

b. Listeria monocytogenes

c. Corynebacterium diphtheriae

d. Erysipelothrix rhusiopathiae

e. Nocardia asteroids

34. หญิงอ้วนอายุ 50 ปี มาห้องฉุกเฉิน มีค่าpH (H) 7.5, PO2 (N) 60, PCO2 (L) 20 และ HCO3- (N) 22, SA O2 (N) 90% มาด้วยอาการผิดปกติอะไร

normoxia

a. combined respiratory and metabolism acidosis

b. acute respiratory acidosis with normoxia

c. acute respiratory alkalosis with hypoxemia

d. combined respiratory and metabolism alkalosis

e. Respiratory alkalosis with compensate



"respiratory alkalosis"

Check ว่า compensate มั้ย?

$pCO_2 = 0.7[HCO_3^-] + 21 \pm 1.5$

$= 0.7(22) + 21 \pm 1.5$

$= 34.9 - 37.9$ mmHg → ไม่ compensate

35. มี air bubble contaminate ใน syringe จะเกิดอะไรขึ้น

air embolism

a. PCO2 เพิ่ม

b. PO2 เพิ่ม

c. H+ เพิ่ม

d. HCO3- เพิ่ม

e. Aa gradientเพิ่ม

36. 80-year-old women มี High grade fever, wet cough for 7 days, Dyspnea, Productive cough, Crepitation at right lung field

(บอก vital sign ต่างๆมา)

Lung biopsy: infiltrate of neutrophil and fibrin into alveolar space

What's the diagnosis?

a. Pneumonia
b. Bronchiectasis
c. TB
d. Emphysema thoraces
e. Acute bronchitis

37. ชายคนหนึ่ง อายุ 7 ปี 1 ปีที่ผ่านมามีอาการ dyspnea และ chronic wet cough ติดเชื้อ pneumonia มาแล้ว 5 ครั้ง ตรวจร่างกายพบ Sat O2 92%, increase AP diameter of chest wall and clubbing of fingers. CXR พบ tram and track opacities at bronchial are และพบ bronchodilatation ถาม pathology?

a. elastin changes of alveoli
b. broncho-arteriole damage
c. bronchial hyperresponsiveness
d. over production of goblet cells
e. destruction of alveoli-capillary basement membrane

38. 15-year-boy was diagnosis post operative right upper lung atelectasis. Positive pressure ventilation applying the patient and follow up CXR show no atelectasis. ถาม mechanism ของการรักษา

positive pressure breathing = อัดอากาศเข้าปอดเลย (ext. intercostal, diaphragm ไม่ต้องทำงาน)

a. increase muscle strength
b. increase transpulmonary pressure
c. decrease work of breathing
d. decrease mean airways pressure
e. more negative in plural pressure

39. 2-years-old infant come to the hospital with coughing while having peanut

- แพทย์สงสัย Aspiration by foreign body

- PE: decreased breath sound at left lung

- CXR (AP) Bilateral purulent at left lobe

(มีภาพ CXR ให้ เจอ homogenous opacity ที่ left lung ติดกับหัวใจ)

What is the cause? ↳ ไม่มี air เข้า

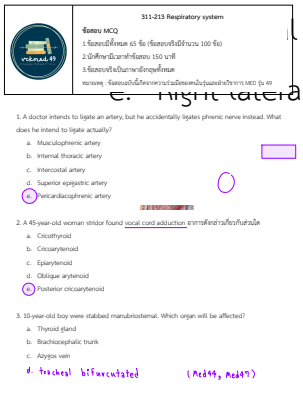
a. Obstruction of carina
b. Partial obstruction at trachea
c. Partial obstruction at main left bronchus
d. Complete obstruction at trachea
e. Complete obstruction at main left bronchus

40. Which lesion obliterate right hemidiaphragm?

a. Pneumothorax
b. A pneumonia in right lower lobe
c. Lung mass in right middle lobe
d. Right breast mass
e. Anterior Mediastinum mass

41. A 30-year-old man with asthma. He had exacerbation and came to hospital. He had dyspnea during admit, then endotracheal tube was inserted. Which CXR technique was used to confirm the location of endotracheal tube?

a. PA
b. Portable, AP view
c. Portable, lateral view
e. Right lateral decubitus


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-Vichakarn Med49-

42. A 32-year-old woman presents with purulent nasal discharge, facial pain, and high-grade fever for 10 years. Plain X-ray shows air-fluid level in both maxillary sinuses. Which of following is the pathophysiology of this condition?

a. Obstruction of paranasal sinuses opening
b. Traumatic fracture of nasal conchae

43. Which is complicated chest injury?

ตอบ Flail chest

44. A 20-year-old man has a history of a recent dental root infection shows up with right facial swelling. A warm, red, tender fluctuant mass push up the root of mouth. What sinus is the most likely to be affected?

a. frontal
b. sphenoid
c. maxillary
d. anterior ethmoid
e. posterior ethmoid

45. Most common site of nasal bleeding

ตอบ Most nosebleeds occur in the anterior part of the nose (Kiesselbach's plexus)

46. Which of the following change occurs during inspiration?

a. Intrapleural pressure is more positive
b. Transpulmonary pressure is more negative
c. Alveolar pressure is lower than atmospheric pressure
d. Lung compliance decrease

47. Which of the following conditions cause the positive pleural pressure?

a. End of forced expiration
b. Beginning of inspiration

c. End of inspiration

d. End of expiration

a. Beginning of expiration

48. Which of the following increase the airway resistance?

a. staying at high altitude

b. in steam room

49. Higher value at apex than base

a. intra alveolar pressure

b. resting volume

50. Condition อะไรส่งผลให้ v/q ratio ลดลง

a. Atelectasis

b. Emphysema

c. Embolism

d. Hemorrhage

51. ทำอะไรถึงทำให้ Pco2 ลดลงในคนสุขภาพดี

a. moderate exercise

b. scuba diving

c. mountain climbing

d. หายใจในถุง

52. Which of the following occurs in the patient with anemia?

a. normal O2 capacity

b. normal PO2 content

c. normal HbO2 saturation

d. decrease arterial PO2

e. decrease HbO2 saturation

Increased intracranial pressure leads to a **rapid and deep breathing pattern**

53. increase intracranial pressure เกิด Breathing pattern แบบไหน

a. Biot's breathing
b. Kussmaul's respiration → Kussmaul breathing is **an abnormal breathing pattern characterized by rapid, deep breathing at a consistent pace**
c. Ondine's curse
d. Apneustic breathing
e. Cheyne-Strokes respiration

54. ชายอายุ 45 ให้อาการ หมอ prescribed chlorpheniramine ถาม moa

ตอบ binds to the histamine H1 receptor. This blocks the action of endogenous histamine, which subsequently leads to temporary relief of the negative symptoms brought on by histamine.

55. 60-year-old woman มีประวัติ น้ำหนักลด 4 กิโลกรัมใน 3 เดือน, hemoptysis, pleuritic chest pain for 2 weeks, CXR เจอ reticulonodular infiltration และ cavitation ถามว่า biopsy จะเจอ histologic finding แบบไหน

a. Septate hyphae fungus with dichotomous branching
b. Epithelioid cell aggregation and Langhans giant cell
c. Squamous epithelioid cells with keratin and numerous mitosis

56. Captain 35 ปี เป็น allergic symptoms ต้องให้ยาอะไรที่สามารถทำงานได้ปกติ

a. Chlorpheniramine
b. Pyroxamine
c. Hydroxyzine

57. Right hilum mass ตรวจยังไง

a. Sputum examination
b. Transthoracic needle biopsy
c. Bronchoscope and biopsy
d. Brushing and cytology
e. Washing and cytology

58. Autopsy พบ goblet cell metaplasia smooth muscle hyperplasia มี infiltrate eosinophil

a. Bronchitis
b. Bronchieectasis
c. Acute bronchitis
d. Chtonic bronchitis
e. Asthma bronchitis

59. A 7-years-old boy had dyspnea and chronic wet cough for 1 year. He was diagnosed pneumonia for 5 times. The physical examination shows SpO2 90%, crackles both lung, AP diameter increase and clubbing fingers. The x-ray reveals tram-track sign peri- bronchial area with bronchial dilatation. What is the pathophysiology of this case?

a. Elastic change of alveoli Not sure Bronchiectasis ?
b. Broncho-alveolar damage
c. Destruction of alveolar-capillary basement membrane
d. Obstruction of mucus production
e. Bronchial hyper responsive

60. A male fetal who born from maternal DM. He develops grunting and cyanosis immediately after birth. His CXR shows bilateral reticulogranular both lungs. Which of the following material is responsible for this condition?

a. mucin
b. lecithin
c. sphingomyelin
d. glucagon-like peptide
e. dipalmitoyl phosphatidylcholine

61. หากต้องการตรวจหาเชื้อจาก saliva ควรปฏิบัติอย่างไรในการเก็บตัวอย่าง

a. งดน้ำหรืออาหารอย่างน้อย 1 ชั่วโมงก่อนเก็บ saliva
b. บ้วนปากด้วยน้ำเปล่าก่อนเก็บตัวอย่าง
c. ดื่มน้ำแล้วจึงเก็บ saliva หลังตื่นนอนในตอนเช้า

62. เด็กพึ่งเกิด ตายในสองวัน มี atelectasis

ตอบ เบาหวาน (gestational diabetes)

63. A 16-year-old lady has acute dyspnea. Her ABG analysis show PH 7.60, PO2 120 mmHg, PCO2 20 mmHg, HCO3- 26 mmol/L, SaO2 99%. What is the interpretation of her ABG analysis?

a. Acute respiratory alkalosis
b. Chronic respiratory alkalosis
c. Metabolic alkalosis, uncompensated
d. Mix respiratory and Metabolic alkalosis
e. Metabolic alkalosis, partially compensated

64. A 60-year-old man, presents with dyspnea on exertion with chronic productive cough for 6 years. He had no fever nor weight loss.

PE: good consciousness, not pale no jaundice, Lung: occasional wheezing and crepitation in both lungs

Heart: no murmur, Extremities: No edema

Which of the following is the possible pulmonary function test result?

a. FEV1/FVC 70% predicted, FVC 65% predicted, TLC 68% predicted
b. FEV1/FVC 85% predicted, FVC 50% predicted, TLC 50% predicted
c. FEV1/FVC 65% predicted, FVC 55% predicted, TLC 95% predicted
d. FEV1/FVC 90% predicted, FVC 30% predicted, TLC 80% predicted
e. FEV1/FVC 50% predicted, FVC 60% predicted, TLC 60% predicted

65. A 40-year-old man current heavy smoking 1 pack per day for 4 years, present with chronic productive cough and dyspnea on exertion for 1 year. What is long term effect of smoking in this patient?

a. Stimulate protease activity
b. Induce high antiprotease activity
c. Increase protease activity
d. Inhibit oxygen free radicals

e. Reduce antitrypsin activity